Mathematical Research Log

Focus: Vertex colouring of a graph

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Daily Insight / Philosophy

- Relative point of view
- Converting an inequality into an equality by finding the 'missing term'.

1.1 Converting the vertex colouring problems into something else

Given a graph Γ[1]. The classical graph theory has many inequality on the chromatic number $\chi(\Gamma)$:

$$\chi(\Gamma) \geq \omega(\Gamma)$$

where $\omega(\Gamma)$ is the clique number of a graph.
Using the degree of a graph $\Delta(\Gamma)$, a graph has an upper bound

$$\chi(\Gamma) \leq \Delta(\Gamma) + 1$$

Hoffman gives a bound [HH70] using spectrum of a graph to get an upper bound of a graph:

$$\chi(\Gamma) \leq 1 - \frac{\lambda_{\max}}{\lambda_{\min}}$$

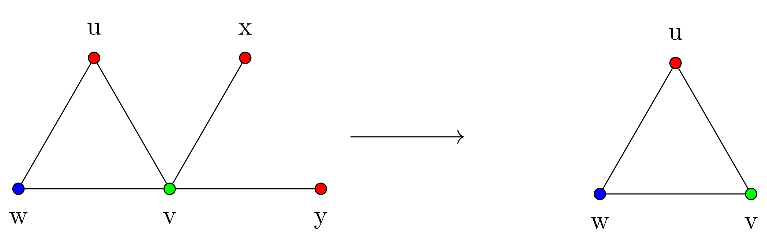
Without doubt, these inequalities are obtained by looking at the graph itself. But, the Grothendieck school taught us the *point du vue relatif*. The first change of perspective is: To study Γ, not looking at Γ itself, but look at the morphisms between Γ and any other graphs.
Fortunately, there is a way to encompass the vertex colouring into a morphism of graphs. Consider the following map for any graph Γ and some integer n:

$$f : \Gamma \to K_n$$

such that $\forall u, v \in V(\Gamma)$ that are incident, $f(u)$ and $f(v)$ are incident in K_n.
In this homomorphism, such an n always exist(n is at most as many as $|V(\Gamma)|$). This is because any graph can be embedded into a complete graph with the number of vertices as many as the number of vertices as Γ. For this n, the existence of an embedding implies that $\chi(\Gamma) \leq \chi(K_n)$. Based on this $n(n = |V(\Gamma)|)$, a homomorphism is still attained by diminishing n.
The problem of finding the (vertex) chromatic number $\chi(\Gamma)$ is converted to find the minimum n such that $f : \Gamma \to K_n$ is a homomorphism of a graph. I would like to rephrase this into the following lemma:

Lemma 1.0.1 (Equivalency of χ and the least complete graph) *Let Γ be a graph. The (vertex) chromatic number $\chi(\Gamma)$ is the minimum number n that $f : \Gamma \to K_n$ is a homomorphism of graphs.*

Proof: □
Here is an example to illustrate the embedding process:



(a) A graph Γ (b) identification of Γ

In this process, one might notice that there is one object that is the 'smallest', which is worthy to being assigned a concept, the **core** of a graph Γ, denoted $\Gamma^\bullet$.
The concept of the core of Γ has already engulfed the embedding. In addition, it is more inclusive than the embedding $f : \Gamma \to K_n$. Because the map $\beta : \Gamma \to \Gamma^\bullet$ allows a section $\varphi : \Gamma^\bullet \to \Gamma$ such that $\beta \circ \varphi = \mathrm{id}_{\Gamma^\bullet}$ at the same time.

Based on the observation on β, there are many streams on the study of β:

[1] Here I use Γ instead of G because G is about to denote a group.

(1.1) β is a retraction map. There are something that relates to a retraction. So, maybe one can study the first fundamental groups between them. A possible question is what is the relation between $\pi_1(\Gamma)$ and $\pi_1(\Gamma^\bullet)$?

(1.2) It might also be possible to extend such a map $\beta : \Gamma \to \Gamma^\bullet$ into a short exact sequence:

(1.2.1)In this case, the χ in the chromatic number might be thought of as a functor from the category of graphs to the category of $\mathbb{Z}$.

(1.2.2)Another thought is that it would be possible to study some groups related to a graph, so then the language and results in group cohomology could be applied:

(1.2.2.1) First, consider the free abelian group generated by the vertex set or by the edge set.

(1.2.2.2) Second, one can consider the group generated by a graph. This is a *directed* way.

(1.2.2.3) Third, there might be groups acting on a graph to reflect the properties of graphs. This is an *indirected* way to study it.

Returning to the number $\chi(\Gamma)$, the existing results are all inequalities. While, I believe all inequalities stem from incomplete observations of structures[2]. In more details, some subtle parts are dismissed and then cause an inequality. I guess there might be two ways to 'complete' this inequalities:

(2.1) The first way is to complete it by adding or subtracting an correction term $\epsilon(\Gamma)$ related to Γ, and then we get

$$\chi(\Gamma) - \epsilon(\Gamma) = \omega(\Gamma)$$

(2.2)

1.2 Technical details

For (1.2), there is a problem to 'extend' $\beta : \Gamma \to \Gamma^\bullet$ into a short exact sequence. Because, β is a morphism in the category of groups, which is not an abelian category. So, it is impossible to talk about the identity and kernel. Let alone the case (1.2.1). The change of point of view is to transfer the category of graphs into some abelian categories by assigning Γ some additional structures. This suggests that (1.2.2) is possible.

1.2.1 Case (1.2.2.1)

(1.2.2.1) One possible way is to consider the free abelian group generated by $V(\Gamma)$ and $E(\Gamma)$, respectively. They are denoted by $C_0(\Gamma)$ and $C_1(\Gamma)$, respectively. More explicitly,

$$C_0(\Gamma) = \left\{ \sum_i a_i v_i \middle| \forall i, v_i \in V(\Gamma), a_i \in \mathbb{Z}, a_i = 0 \text{ cofinitely} \right\}$$

Similarly notation for $C_1(\Gamma)$. Their relations are due to a boundary map ∂,

$$\partial : C_1(\Gamma) \to C_0(\Gamma) \qquad e \mapsto \psi(e) - \phi(e)$$

where Γ has an orientation E^+ and $\phi(e)$ is the start and $\psi(e)$ is the end of e.

1.2.2 Some variants for (1.2.2.1): Lovász's theory

1.2.3 Case (1.2.2.2)

(1.2.2.2)

[2]Algebraists tend to deal with equalities and isomorphisms. I gained such a point of view from them.

1.2.4 Case (1.2.2.3)

To-Do List

- ☐ Tutte Polynomial ⇒ Bollobás-Riordan Polynomial (2000) for Ribbon Graph
- ☐ Graph Motives
- ☐ Chromatic symmetric function: Richard Stanley (1995) introduced X_Γ. Vertex-coloring problem ⇒ Schur functions+ representation theory.
- ☐ Hom-complex: What is the topological degree of $\mathrm{Hom}(\Gamma, K_n)$? What is the homology group of $\mathrm{Hom}(\Gamma, K_n)$?
- ☐ Vertex-coloring ⇒ vector-bundles

To-Do List

Electricity Graphs (Flows):

Mapping Kirchhoff's laws (Voltage Current) to the language of Homology groups.

Utilizing Universal Coverings to analyze flows and circuits within complex loops.

To-Do List

Euler's Theory on Graphs:

Studying the fundamental groups of graphs embedded in various surfaces.

Exploring dual graphs and the application of Poincaré duality in a graph-theoretic context.

To-Do List

Witten's Perspective: Exploring the connection between chromatic polynomials, knot polynomials, and statistical physics (e.g., the Potts model).

June Huh's Work: Understanding the intersection of Algebraic Geometry (AG) and Graph Theory, particularly regarding the log-concavity of chromatic polynomials.

Topological Constraints(Planarity and Vertex-coloring): Relative perspectives on chromatic polynomials in embeddings and proofs involving Kuratowski's theorem (K_5 and $K_{3,3}$).

References

[HH70] A. J. Hoffman and Leonard Howes. "ON EIGENVALUES AND COLORINGS OF GRAPHS, II". In: *Annals of the New York Academy of Sciences* 175.1 (1970), pp. 238–242. DOI: https://doi.org/10.1111/j.1749-6632.1970.tb56474.x. eprint: https://nyaspubs.onlinelibrary.wiley.com/doi/pdf/10.1111/j.1749-6632.1970.tb56474.x. URL: https://nyaspubs.onlinelibrary.wiley.com/doi/abs/10.1111/j.1749-6632.1970.tb56474.x.